

Answer all questions.

1. (a) The following table shows some information about Group 1 elements.

Metal	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Reaction with chlorine
lithium	180	1342	0.54	reacts slowly to make a white salt
sodium	97	883	0.97	burns vigorously with a yellow flame to make a white salt
potassium	63	759	0.88	reacts violently to make a white salt
rubidium	39	688	1.53	explosive reaction
caesium	28	671	1.93	explosive reaction

- (i) Describe the trend in density going down the group.

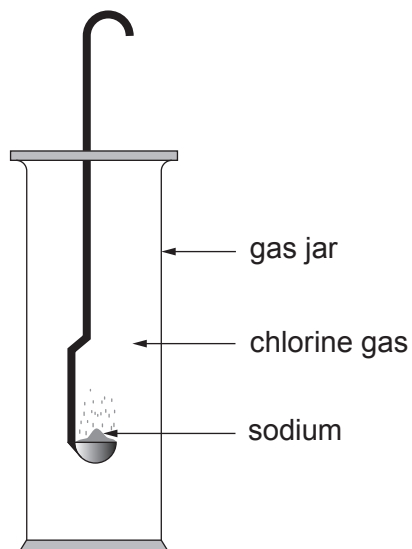
[1]

- (ii) Explain the difference in reactivity down the group in terms of electronic structure.

[2]



- (b) The apparatus below can be used to demonstrate the reaction between sodium and chlorine.



- (i) Apart from the use of safety goggles, state **one** safety precaution that needs to be followed when using **each** of these elements. [2]

Element	Safety precaution
sodium	
chlorine	

- (ii) Complete and balance the symbol equation for the reaction that takes place between sodium and chlorine. [2]

